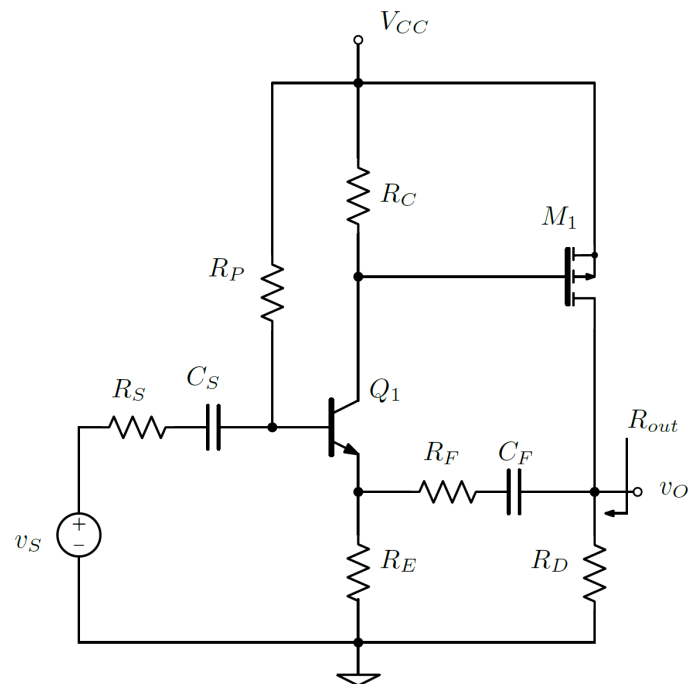


E1



Consider the two stage amplifier in the figure, whose parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 12\text{ V}$; $R_S = 1\text{ k}\Omega$; $R_P = 1.2\text{ M}\Omega$; $R_C = 10\text{ k}\Omega$; $R_E = 220\text{ }\Omega$; $R_F = 1.2\text{ k}\Omega$; $R_D = 2.2\text{ k}\Omega$; $C_S = 1\text{ }\mu\text{F}$; $C_F = 220\text{ }\mu\text{F}$;

Q1 (npn BJT): $V_{BE} = 0.7\text{ V}$; $\beta_F = \beta_0 = 100$; $r_o = +\infty$;

M1 (e-PMOS): $V_t = -3\text{ V}$; $I_{DSS} = 0.3\text{ mA}$; $r_o = +\infty$.

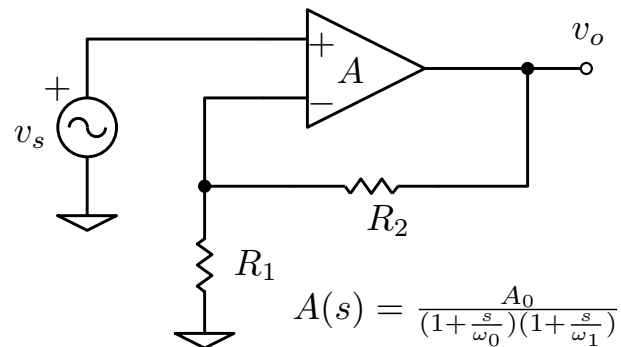
Determine:

1. the bias point of Q_1 and Q_2 (voltage and current on the devices);
2. the equivalent small signal circuit according to feedback theory, indicating the type of sampling and mixing, plus the equivalent, two-port model of the beta-network;
3. the closed loop gain $A_F = v_o/v_s$;

Optionally, also determine the following parameters:

5. the output resistance indicated in the figure (closed loop);
6. an estimation of the amplifier *low frequency cut-off* ω_L (closed loop), assuming a dominant pole exists in the low frequency response of the amplifier.

E2



Consider the non inverting amplifier shown in the figure. Assuming the operational amplifier is just partially compensated with $A_0 = 10^5$ [V/V], $\omega_0 = 10$ krad/s, $\omega_1 = 100$ krad/s, determine:

1. the estimated phase margin when $R_1 = 10\text{k}\Omega$ and $R_2 = 330\text{k}\Omega$;
2. in case dominant pole compensation is used, the compensation network parameters that correspond to a dominant pole location at $\omega = \omega_{\text{DP}} = 660$ krad/s;
3. the circuit schematic including the compensation network;
4. the estimated bandwidth and phase margin after compensation.